

ONLINE TENDER MANAGEMENT SYSTEM

Team Members

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1. INTRODUCTION

The **Online Tender Management System** is a web-based application that manages the complete tender process online.

An **Organization/Tender Authority** can create and publish a tender with project requirements, budget, deadline, eligibility criteria, and evaluation criteria. Registered **Companies/Bidders** can view suitable tenders and submit their bids along with technical proposals, project plans, quotations, and supporting documents.

After the bidding deadline, the organization that created the tender evaluates all submitted bids and selects the best bidder.

The system has three main users:

- **Administrator** – manages the overall platform.
 - **Organization/Tender Authority** – creates tenders and selects the winning bid.
 - **Company/Bidder** – searches tenders and submits bids.
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2. PROBLEM STATEMENT

Traditional tender management involves paperwork, manual bid collection, document verification, and difficult comparison of proposals.

This can lead to:

- Time consumption
- Large amounts of paperwork
- Difficulty managing documents
- Manual errors

- Difficulty comparing bids
- Delayed communication
- Difficulty tracking tender status

Therefore, an online system is proposed to make the tender process faster, organized, and easier to manage.

3. OBJECTIVES

The main objectives of the Online Tender Management System are:

1. To provide an online platform for tender management.
 2. To allow organizations to create and publish tenders.
 3. To allow companies to search and participate in tenders.
 4. To allow companies to submit bids and supporting documents.
 5. To allow organizations to evaluate and compare bids.
 6. To allow the organization to select the best bidder.
 7. To provide notifications about important tender events.
 8. To maintain all tender and bid information in a centralized database.
 9. To reduce paperwork and manual effort.
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4. SCOPE OF THE SYSTEM

The system covers the following activities:

- User registration and login
- Organization management
- Company management
- Tender creation
- Tender publication
- Tender searching
- Bid submission
- Document upload

- Bid evaluation
- Bid comparison
- Winner selection
- Tender award
- Notifications
- Tender history

The system can later be extended with online payments, digital signatures, mobile applications, and AI-assisted evaluation.

5. USERS AND THEIR RESPONSIBILITIES

Administrator

The Administrator manages the overall platform.

Responsibilities:

- Manage users
- Manage organizations and companies
- Approve or deactivate accounts
- Monitor the system
- View reports

Organization / Tender Authority

The Organization/Tender Authority is the organization that creates and publishes a tender.

Responsibilities:

- Create tenders
- Define requirements
- Set budget and deadline
- Publish tenders
- View submitted bids
- Evaluate bids
- Compare bids

- Select the best bidder
- Award the tender

Company / Bidder

The Company/Bidder is a company that participates in tenders.

Responsibilities:

- Register and maintain company profile
 - Search available tenders
 - Check eligibility
 - Submit bids
 - Upload technical proposals
 - Submit financial quotations
 - Upload supporting documents
 - Track bid status
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6. MAIN FUNCTIONAL REQUIREMENTS

The system should provide the following functionalities:

- User registration and login
- Role-based access
- Organization profile management
- Company profile management
- Tender creation and publication
- Tender search and filtering
- Eligibility checking
- Bid submission
- Technical proposal submission
- Financial quotation submission
- Document upload
- Bid evaluation

- Bid comparison
- Winner selection
- Tender award
- Notifications
- Tender status tracking

7. TENDER WORKFLOW

The complete workflow of the Online Tender Management System is:

ORGANIZATION

|

v

Create Tender

|

v

Define Requirements

|

v

Set Budget & Deadline

|

v

Publish Tender

|

v

+-----+

| Available Tender |

+-----+

/ | \

/ | \

v v v

Company A Company B Company C

| | |

v v v

Bid A Bid B Bid C

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\ | /

\ | /

v

Bidding Closes

|

v

Organization Reviews

|

v

Evaluate Bids

|

v

Compare Bids

|

v

Select Best Bid

|

v

Award Tender

|

v

Notify Companies

Important: The **organization that created the tender** selects the winning company.
The Administrator does not choose the winner.

8. TENDER AND BID DETAILS

Tender Details

When an organization creates a tender, it provides:

- Tender ID
- Tender title
- Description
- Requirements
- Category
- Estimated budget
- Start date
- Submission deadline
- Eligibility criteria
- Required documents
- Evaluation criteria
- Terms and conditions
- Tender status

Bid Details

When a company submits a bid, it provides:

- Bid ID
- Company details
- Proposed price
- Technical proposal
- Project plan
- Completion time
- Previous experience

- Supporting documents
- Submission date

9. BID EVALUATION

Bid evaluation is performed by the organization that created the tender.

The organization can define evaluation criteria before publishing the tender.

For example:

Evaluation Criterion Weight

Technical Proposal 40%

Company Experience 20%

Financial Proposal 25%

Completion Time 10%

Documentation 5%

Total 100%

The organization evaluates each submitted bid according to these criteria.

The system can store:

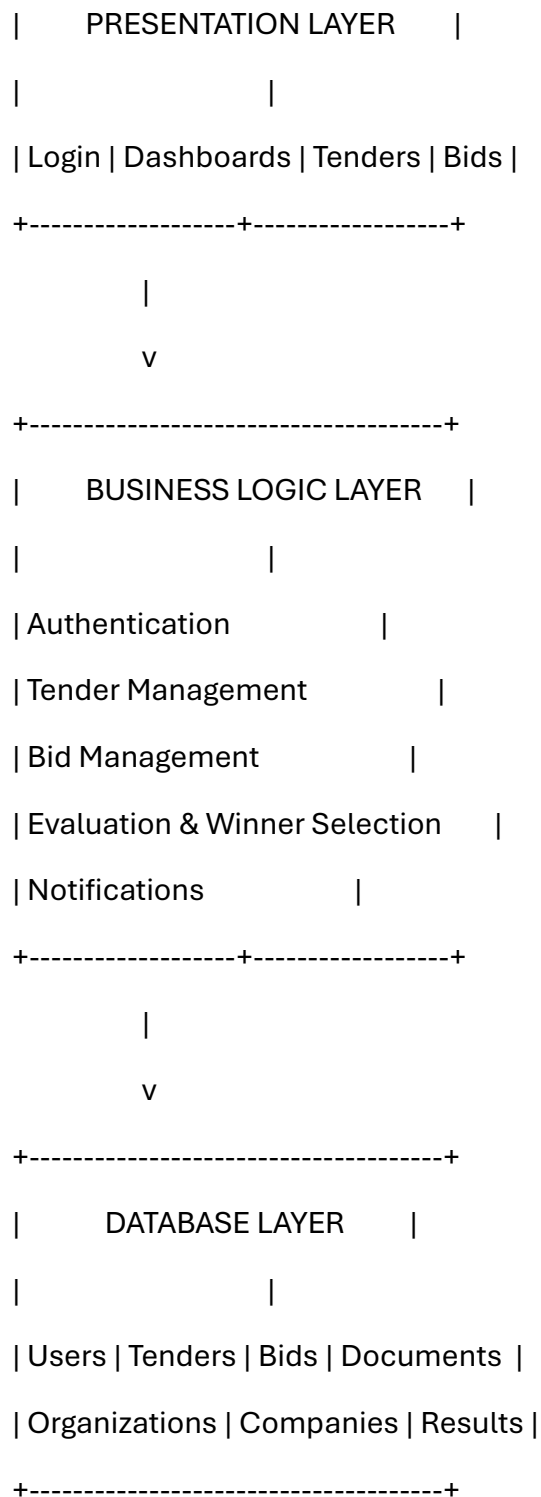
- Evaluation scores
- Total score
- Organization remarks
- Selected bidder
- Award status

The organization compares the bids and selects the company providing the best overall proposal.

10. SYSTEM ARCHITECTURE

The system follows a **three-tier architecture**.

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Presentation Layer

Provides the user interface for:

- Login
- Registration
- Dashboards

- Tender browsing
- Bid submission
- Bid evaluation
- Reports

Business Logic Layer

Handles:

- Authentication
- Authorization
- Tender management
- Bid management
- Eligibility validation
- Evaluation
- Winner selection
- Notifications

Database Layer

Stores:

- Users
- Organizations
- Companies
- Tenders
- Bids
- Documents
- Evaluations
- Notifications

11. DATABASE DESIGN

USER

User

user_id (PK)

name

email

password

role

status

ORGANIZATION

Organization

organization_id (PK)

name

email

phone

address

registration_no

status

COMPANY

Company

company_id (PK)

company_name

email

phone

address

registration_no

experience

status

TENDER

Tender

tender_id (PK)

organization_id (FK)

title

description

budget

start_date

deadline

eligibility

evaluation_criteria

status

BID

Bid

bid_id (PK)

tender_id (FK)

company_id (FK)

bid_amount

technical_proposal

project_plan

completion_time

submission_date

status

DOCUMENT

Document

document_id (PK)

bid_id (FK)

document_name

document_path

upload_date

EVALUATION

Evaluation

evaluation_id (PK)

bid_id (FK)

technical_score

experience_score

financial_score

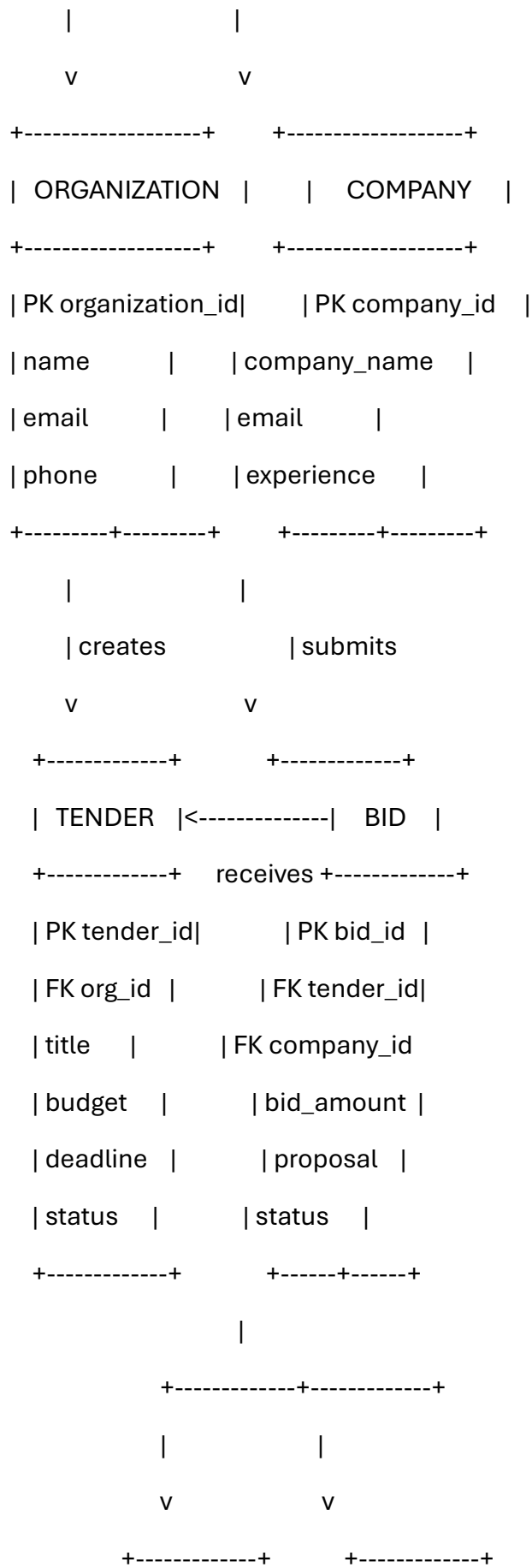
time_score

total_score

remarks

12. DATABASE / ER DIAGRAM

```
+-----+
|  USER  |
+-----+
| PK user_id |
| name      |
| email     |
| role      |
+-----+-----+
|
+-----+-----+
```



DOCUMENT		EVALUATION	
+-----+		+-----+	
document_id		evaluation_id	
bid_id		bid_id	
name		scores	
path		total_score	
+-----+		remarks	
+-----+			

Important Relationships

Organization 1 ---- N Tender

Tender 1 ---- N Bid

Company 1 ---- N Bid

Bid 1 ---- N Document

Bid 1 ---- 1 Evaluation

This means:

- One organization can create many tenders.
- One tender can receive many bids.
- One company can submit many bids.
- One bid can contain multiple documents.
- One bid has one evaluation.

13. NON-FUNCTIONAL REQUIREMENTS

Security

Only authorized users should access restricted information.

Performance

The system should respond quickly to normal user requests.

Usability

The interface should be simple and easy to understand.

Reliability

Tender and bid information should be stored accurately.

Scalability

The system should support an increasing number of users, organizations, companies, tenders, and bids.

Maintainability

The system should be easy to update and maintain.

Data Privacy

A company must not be able to view another company's submitted bid, quotation, or documents.

Only authorized users from the organization that owns the tender can access the submitted bids.

14. FUTURE ENHANCEMENTS

The system can be enhanced in the future with:

Online Payment

Organizations can collect tender fees and security deposits online.

Digital Signature

Digital signatures can be integrated for important tender documents.

Advanced Encryption

Sensitive bids and documents can be encrypted for better security.

Mobile Application

Android and iOS applications can be developed.

AI-Assisted Evaluation

AI can help summarize proposals and assist organizations in comparing bids. The final decision remains with the authorized organization.

Email and SMS Notifications

Important tender updates can be sent through email and SMS.

Blockchain

Blockchain can be used to create tamper-resistant tender records.

15. CONCLUSION AND TEAM DIVISION

Conclusion

The **Online Tender Management System** provides a centralized platform for managing the complete tender process.

Organizations can create and publish tenders, while companies can search for tenders and submit proposals, quotations, and supporting documents. After the bidding period ends, the organization evaluates and compares the bids and selects the best bidder.

The system reduces paperwork, saves time, improves document management, simplifies bid comparison, and provides better tracking of the tender lifecycle.

The Administrator manages the overall platform, organizations manage their own tenders, and companies participate by submitting bids.

Thus, the proposed system provides an efficient, organized, and scalable solution for managing tenders digitally.

Team Division

Member 1

- **Project setup**
- **Authentication and Authorization**
- **Tender management**
- **Bid submission**
- **Evaluation module**
- **Security**

Member 2

- **Document management**
- **Reports and Analytics**
- **Dashboard**
- **AI module**

Member 3

- **User Management**
- **Organization Management**
- **Vendor Management**
- **Notification Module**
- **Deployment**
- **Testing**